EYE-TECH

BUILDING THE DIGITAL EYE

ET2 - GS-FC

Spec document

CONFIDENTIAL

Authors	Monica Vatteroni
ID Project Name	EYE2DRIVE
Date	16 September 2025
Revision	03
Status	Pending
Approved by	
Number of Pages	24

Scope / Overview	Specification of ET2-GS - FCchip

Last revision	00	Data	23/04/2025

Record of revisions	Rev Data	Rev n°	Notes
	23/04/2025	00	First release
	04/06/2025	01	Update of Fig. 3.1, Tab. 3.1 pin list at section 4.1
	06/08/2025	02	Document updated after design block definition
	16-09-2025	03	Update document's name and sensor name

+

CONFIDENTIAL

Index

Index 3

1. Introduction 5

2. Key Features 5

3. Architecture specification 6

4. Pin Configuration 8

4.1. Pin list 8

4.1.1. PinOut 10

4.1.2. Die Outline Dimensions and Mechanical Tolerances 11

4.2. Package mechanical outline 11

5. Operation and timing of the Camera 12

5.1. Read-out timing 13

5.1.1. Pixel Array 17

5.1.2. Video Line 19

5.1.3. Fixed Pattern Noise 19

5.1.4. Decoders 19

5.1.5. Column ADC 20

5.1.6. Analog references 20

6. EYE-TECH Digital core 21

6.1. Operating modes 21

6.2. I2C 22

6.2.1. Communication Protocol 22

6.2.2. Register Map 23

6.2.3. Controller 25

6.2.4. Top-Level Architecture 26

6.2.5. Integration time setting 27

6.2.6. Sub-windowing and sub-sampling modality 28

7. Column ADC 34

8. Output interface – MIPI CSI-2 Tx D-PHY 35

8.1. MIPI Input Interface 39

8.2. CSI-2 Controller 39

8.3. D-PHY TX 42

9. Integration reference 43

10. Appendix A – data timing 44

10.1. Integration/Readout Phase of the ET2-GS-FC Sensor 44

11. Appendix B – design rules notes 46

11.1. Notes for IO library: 46

11.2. Analog power rooting 46

11.3. Digital PADs 48

11.4. Analog PADs 50

11.5. CUT cells 50

1. Introduction

The **ET2-GS-FC** is a monolithic system-on-chip image sensor featuring a **2Mpixel** resolution (1920x1080) and a **2/3-inch** optical format, designed with a global shutter architecture. It integrates EYE2DRIVE technology to deliver high dynamic range performance within a single frame, effectively handling unpredictable lighting conditions without the need for post-processing reconstruction.

This sensor represents a pre-production prototype, engineered to support early-stage development of imaging systems. It includes a digital **2-wire input interface** (I2C-like) and supports image data streaming through a **4-lane MIPI-CSI-2** serial digital output.

Sensor control and the generation of internal reference voltages and currents are fully integrated on-chip, simplifying system design and reducing external component requirements.

2. Key Features

- 1920x1080-pixel resolution,
- 4.8um pixel pitch with global shutter
- 2/3inch, Ø 11 mm optical format
- 16:9 aspect ratio
- Tuneable dynamic range through patented technology (65-120dB)
- 45 fps @full frame
- 4-lane MIPI-CSI-2
- 3.3V/1.8V operating voltage
- Colour and B&W
- 4e- noise flow
- 110uV/e- Conversion gain
- 8.5 ke- Full Well Capacitance

3. Architecture specification

The architecture of the ET2-GS-FC chip has been developed with a clear focus on production readiness, ensuring a smooth transition to volume manufacturing. The sensor features a standard image format of 1920x1080 pixels with a 2/3-inch optical format, corresponding to an 11 mm diagonal.

Sensor configuration is managed through a custom 2-wire I2C-like interface, which communicates with an on-chip controller responsible for orchestrating all internal operations. This includes synchronization and addressing of the pixel array, output data handling, and the configuration of internal reference voltages and currents.

The pixel array readout is performed through 1920 column-parallel ADCs, whose outputs are serialized and transmitted via a 4-lane MIPI-CSI-2 interface.

Additional functional blocks integrated on the chip include array address decoders, a reference voltage and current generation block, PLL, power-on reset (POR) circuitry, and a set of I/O pads for signal interfacing, the exact configuration of which will be defined in later stages of development.

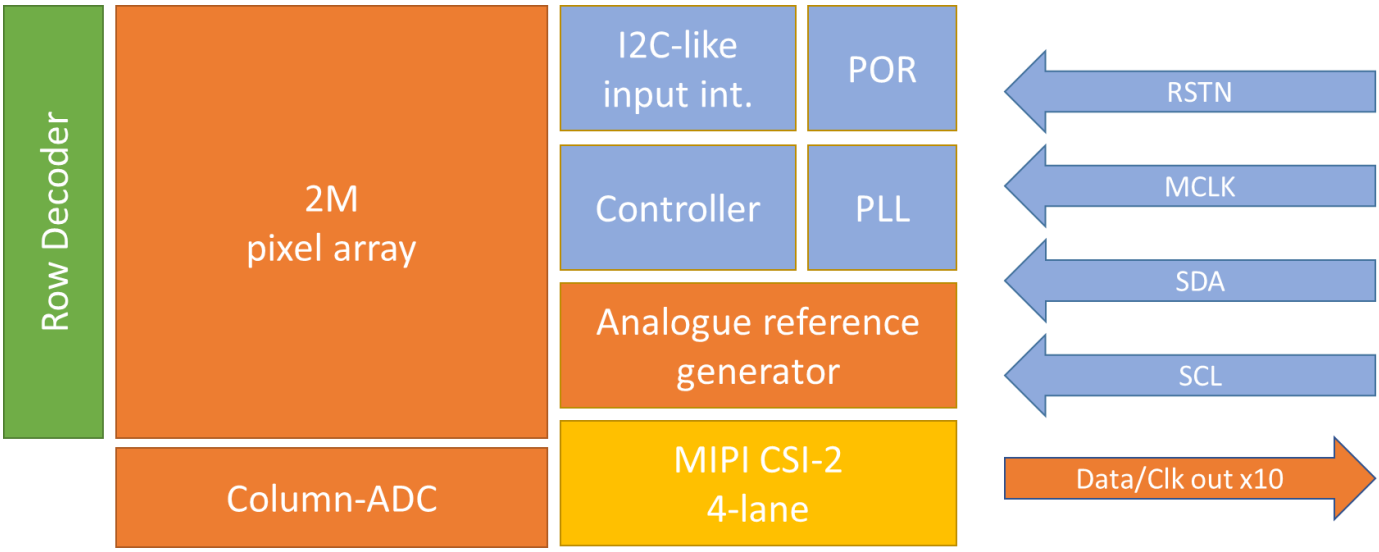


Fig. 3.1 Basic sensor architecture

Tab. 3.1 IP blocks and ownership

IP name	IP description	Ownership
Pixel	4.8um pitch 6T pinned photodiode	Tower
Decoders	Row selection, reset and signal transfer decoders	Monica

Column ADCs	10-bit column ADC	Michel Moser
Charge pumps	-0.5V and 3.5V charge pumps	Lorenzo Dati
MIPI D-PHY Tx	4-lane 2Gps MIPI D-PHY Tx including CSI-2 controller	Lorenzo D and Lorenzo B
I2C-like interface	Input interface	Lorenzo B/RAME
Digital controller	Sensor Timing generation, including test modes	Internal
Bandgap	1.2V Low Noise Bandgap	RAME
POR	Power-on reset circuit for 1.8V (digital blocks power up)	RAME
PLL	500 MHz phase locked loop	RAME
YFlash	FTP 3.3V 64b	Tower
FS191	Standard cell libraries 1.8V Std -40C-150C	Tower
Analogue reference generator	Generation of Current and Voltage reference for the full chip starting from band gap reference (1.2V)	???
Reference Buffer	High load/Low bandwidth buffer for reference voltages	???
DAC	8-bit DAC for GRST internal generation	???

4. Pin Configuration

4.1. Pin list

The ET2-GS-FC sensor pin list is reported in the following table:

Pin	Pin name	Dir	Type	Description	PAD	
DIGITAL						
1	MCLK	I	D	External master clock	PIOH_33_18	
2	RSTN	I	D	Global software reset of the system	PIOH_33_18	
	START	I	D	Global start	PIOH_33_18	
3	SCL	O	D		PIOH_33_18	
4	SDL	I/O	D		PIOH_33_18	
5	SIGN_EN	I	D	Enable of the automatic (low) or manual (high) configuration	PIOH_33_18	
6	TEST_MODE	I	D	Scan chain mode input	PIOH_33_18	
7	TEST_SE	I	D	Scan chain select input	PIOH_33_18	
8	VSYNC	O	D	Frame sink signal	PIOH_33_18	
9	HSYNC	O	D	Line Sink signal	PIOH_33_18	
11	Dec_D	I	D	Global decoder data input	PIOH_33_18	
12	Ex_RowTx2_EN	I	D	Row Tx2 decoder enable	PIOH_33_18	
13	Ex_RowSEL_EN	I	D	Row SELECT decoder enable	PIOH_33_18	
14	Ex_RowRES_EN	I	D	Row RESET decoder enable	PIOH_33_18	
17	Ex_Dec_S0	I	D	Global relet to 0	PIOH_33_18	
18	Ex_RowTx2_FT0	I	D	Row Tx2 global mask to 0 signal	PIOH_33_18	
19	Ex_RowSEL_FT0	I	D	Row SELECT global mask to 0 signal	PIOH_33_18	
20	Ex_RowRES_FT0	I	D	Row RESET global mask to 0 signal	PIOH_33_18	
21	RowTx2_Q	O	D	Row Tx2 decoder data output signal	PIOH_33_18	
22	RowSEL_Q	O	D	Row SELECT decoder data output signal	PIOH_33_18	

23	RowRES_Q	O	D	Row RESET decoder data output signal	PIOH_33_18	
	Ex_Tx1	I	D	External Global transfer signal generation	PIOH_33_18	
24	MIPI_Clk_P	O	D	Differential high-speed clock for D-PHY – P	PAH_33_NA	
25	MIPI_Clk_N	O	D	Differential high-speed clock for D-PHY – N	PAH_33_NA	
26	MIPI_HS_Data_Lane0_P	O	D	Differential high-speed data for lane 0 – P	PAH_33_NA	
27	MIPI_HS_Data_Lane0_N	O	D	Differential high-speed data for lane 0 – N	PAH_33_NA	
28	MIPI_HS_Data_Lane1_P	O	D	Differential high-speed data for lane 1 – P	PAH_33_NA	
29	MIPI_HS_Data_Lane1_N	O	D	Differential high-speed data for lane 1 – N	PAH_33_NA	
30	MIPI_HS_Data_Lane2_P	O	D	Differential high-speed data fo lane 2 – P	PAH_33_NA	
31	MIPI_HS_Data_Lane2_N	O	D	Differential high-speed data for lane 2 – N	PAH_33_NA	
32	MIPI_HS_Data_Lane3_P	O	D	Differential high-speed data for lane 3 – P	PAH_33_NA	
33	MIPI_HS_Data_Lane3_N	O	D	Differential high-speed data for lane 3 – N	PAH_33_NA	
1	PixSupp	S	P-A	Pixel Array supply voltage (3.0V)	PAH_33_NA	
2	Ex_GRST	S	P-A	External Global blooming and dynamic control (discharge ramp 0.0V-3.3V)	PAH_33_NA	
3	VDDA	S	P-A	0.0V Analog power supply	PVDI_NA_18	
4	GNDA	S	P-A	3.3V Analog power supply	PV0I_33_18	

5	VDDH	S	P-A	3.5V power supply used as high level for RES and SEL row signals	PAL_NA_18	
6	VSSH	S	P-A	-0.5V low level power supply used for low level Tx2 signal	PAL_NA_18	
1	GND	S	P-A	0.0V Digital power supply	PV0I_33_18	
2	VDD	S	P-A	1.8V Digital power supply	PVDI_NA_18	
3	GNDA	S	P-A	0.0V Analog power supply	PV0I_33_18	
4	VDDA	S	P-A	3.3V Analog power supply	PVDI_NA_18	
5	VDDO	S	P-P	3.3V PAD Ring power supply	PVDA_33_NA	
6	VSSO	S	P-P	0.0V PAD Ring power supply	PV0A_33_18	

Key:	I	Input	D	Digital Pin with programmable drive strength
	O	Output	A	Analog Pin
	I/O	Input / Output	P-P	Power supply for pad output buffers
	S	Power Supply	P-D	Power supply for core logic
			P-A	Power supply for analog core

All ground and power supply pins must be connected to the proper value and not left floating.

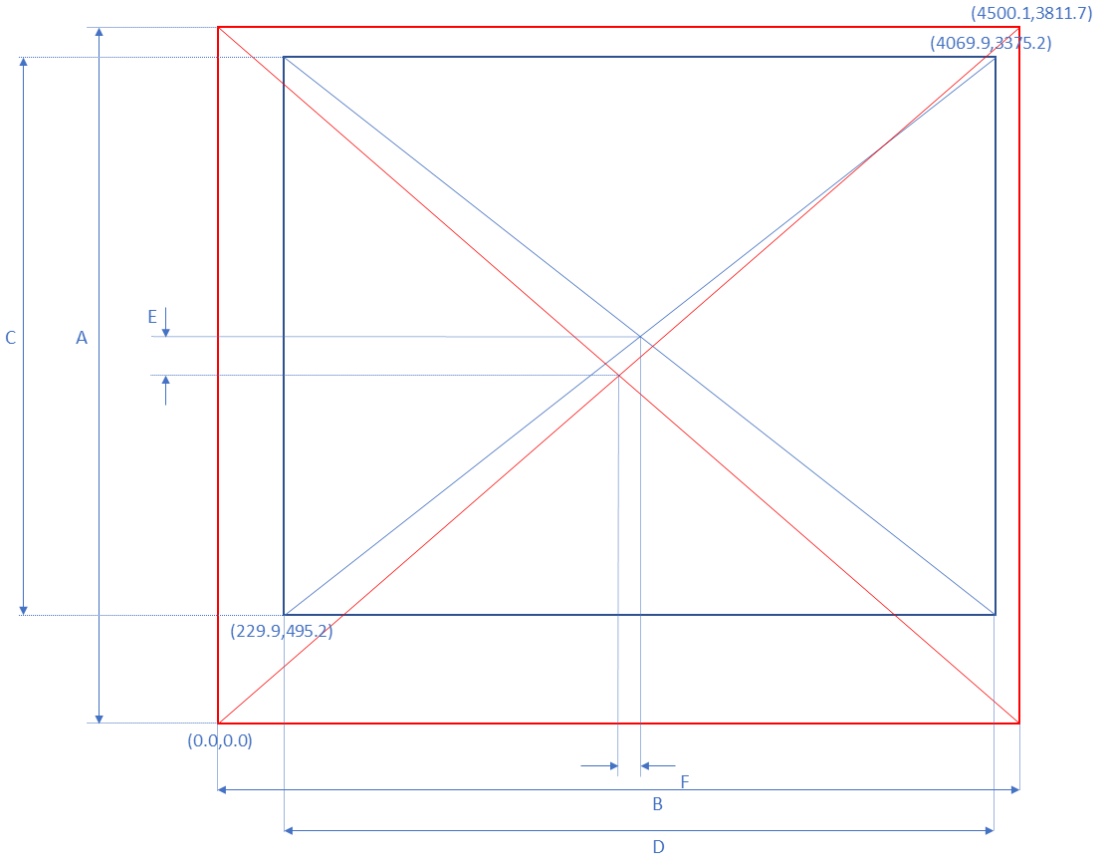
Sinking current flows from VDDA through the bias generator to the pin.

Sourcing current flows from the pin through the bias generator to GNDA.

4.1.1. PinOut

Fig. 4.1 Chip pinout

4.1.2. Die Outline Dimensions and Mechanical Tolerances



Reference	Note	Typ (µm)
A	Y-size of the die	
X	X-size of the die	
C	Y-size of the sensor array	
D	X-size of the sensor array	
E	Y-offset between sensor array center and die center	
F	X-offset between sensor array center and die center	
	Die thickness	297.0

4.2. Package mechanical outline

Fig. 4.2 LCC04811 mechanical case outline

5. Operation and timing of the Camera

All functions of the **ET2-GS-FC** chip are controllable both internally (by a dedicated on-chip controller) and externally, to maximize system flexibility.

After a complete reset of the pixel array's photodiodes through a pulse on the GRST signal, the entire array enters the integration phase for a period determined by the user (the integration time).

During integration, the discharge of the photodiodes is modulated by **a piece-wise-linear signal applied globally to the GRST** gate. This conditioning signal enables controlled discharge across the array, resulting in an output with an enhanced dynamic range depending on the signal shape.

At the end of the integration period, the photodiode signals are globally transferred to the SN junctions by means of a Tx1 global pulse, making the signals ready for readout.

The pixel signals are then transferred row by row to the column ADCs for external readout.

Before reading each signal by enabling Tx2, the output stage of the selected row is reset through an RST pulse, and the reset levels are sampled by the corresponding column ADCs. After the reset sampling, RST is released, and Tx2 is enabled to allow the light signal to be transferred to the column readout stage. The column ADCs then perform correlated double sampling (CDS), subtracting the reset value from the signal value, before digitizing the data for output.

The operations of Tx2, RST, and the ADC double sampling are managed by the RowTx2, RowRES, and ColRES decoders. More details on these decoders and on the timing for generating the correct signal sequence are provided in Sections 5.1.4 and 7.

Once a row's analog signals are digitized, the resulting pixel data are temporarily stored in a row buffer. This buffer sequentially feeds the pixel data into a serializer, which converts the parallel data into a serial stream. The serialized data are then formatted into CSI-2 compliant packets by the MIPI CSI-2 transmitter. These packets include necessary headers, payloads, and error-checking codes, and are transmitted over differential high-speed lanes to the receiving processor or system (more details in section 7).

Each pixel undergoes a double readout sequence (first the signal, then the reset), both synchronized with the column ADC selection and reset operations.

Further details about the pixel structure and decoder behavior are discussed in Sections 5.1.1 and 5.1.3, respectively.

It is important to underline that the global shutter operation allows integration and readout to occur in parallel: a new integration phase can start immediately after the Tx1 signal is closed following the signal transfer, thereby optimizing the sensor's frame rate performance.

A flowchart of the sensor operation is shown in Figure 5.3.

5.1. Read-out timing

The output data frequenct is 2Gbps over 4 differential lines (500MHz each) that, potentially can manage over 60fps.

Supposing a frame rate of 45 fps, the samples per second are 0.93 Gbps and the maximum integration time is 22msec. Row operations must be completed with a frequency of 0.86 MHz and means 1.16 usec per row. This is also the total operation and conversion time of the Column ADC (supposing a single row).

Supposing a frame rate of 60 fps, the samples per second are 1.24 Gbps and the maximum integration time is 16.7msec. Row operations must be completed with a frequency of 1.15 MHz and means 0.87 usec per row. This is also the total operation and conversion time of the Column ADC (supposing a single row).

Fig. 5.1Integration and pixel sampling

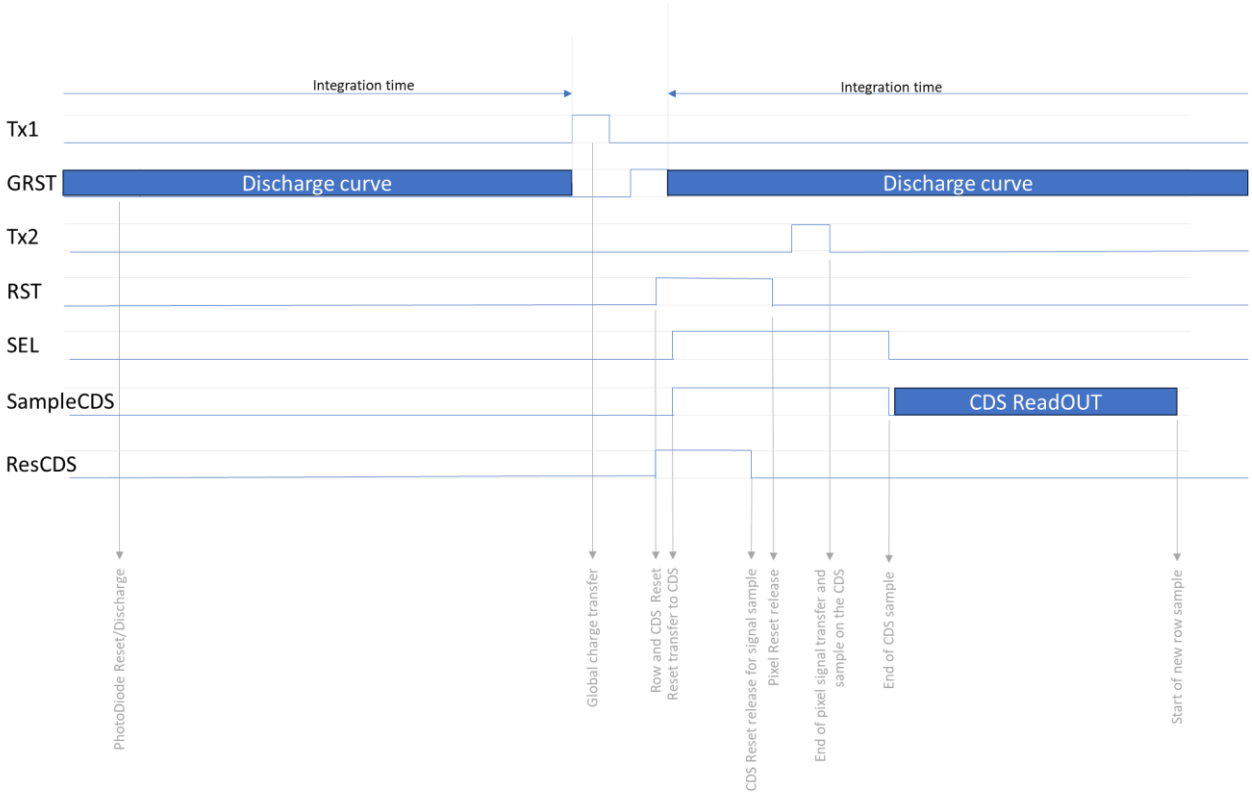
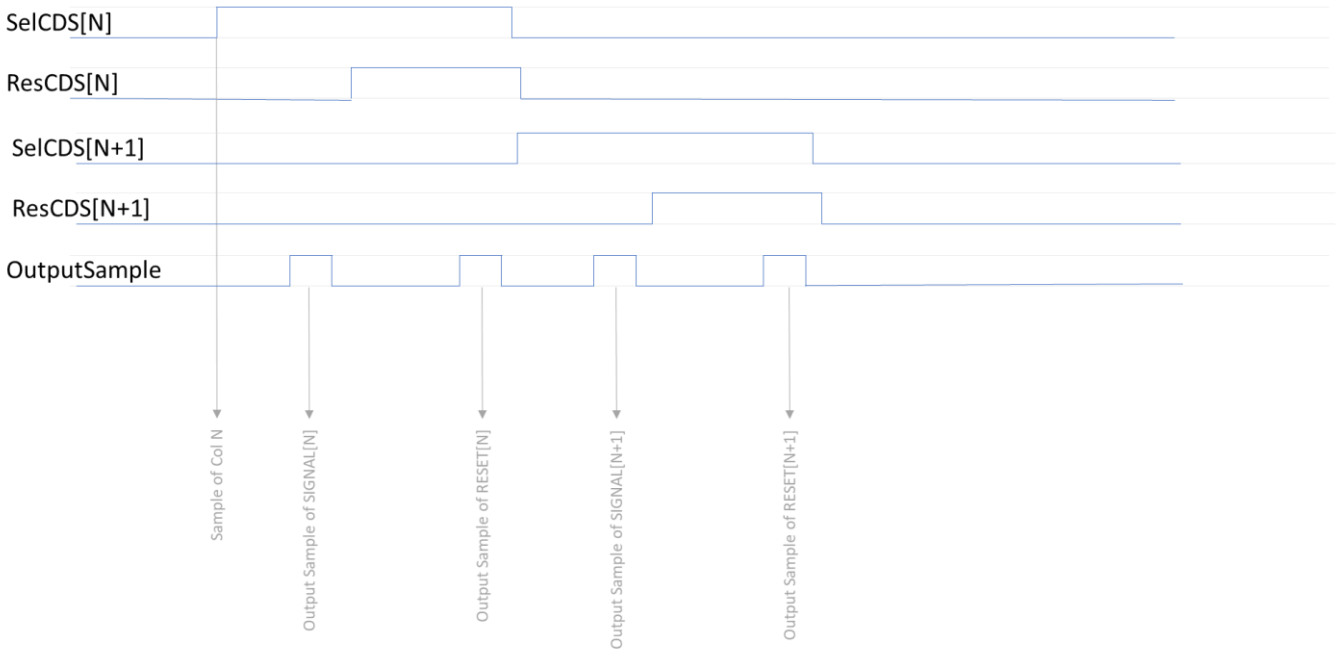


Fig. 5.2 Columns' readout



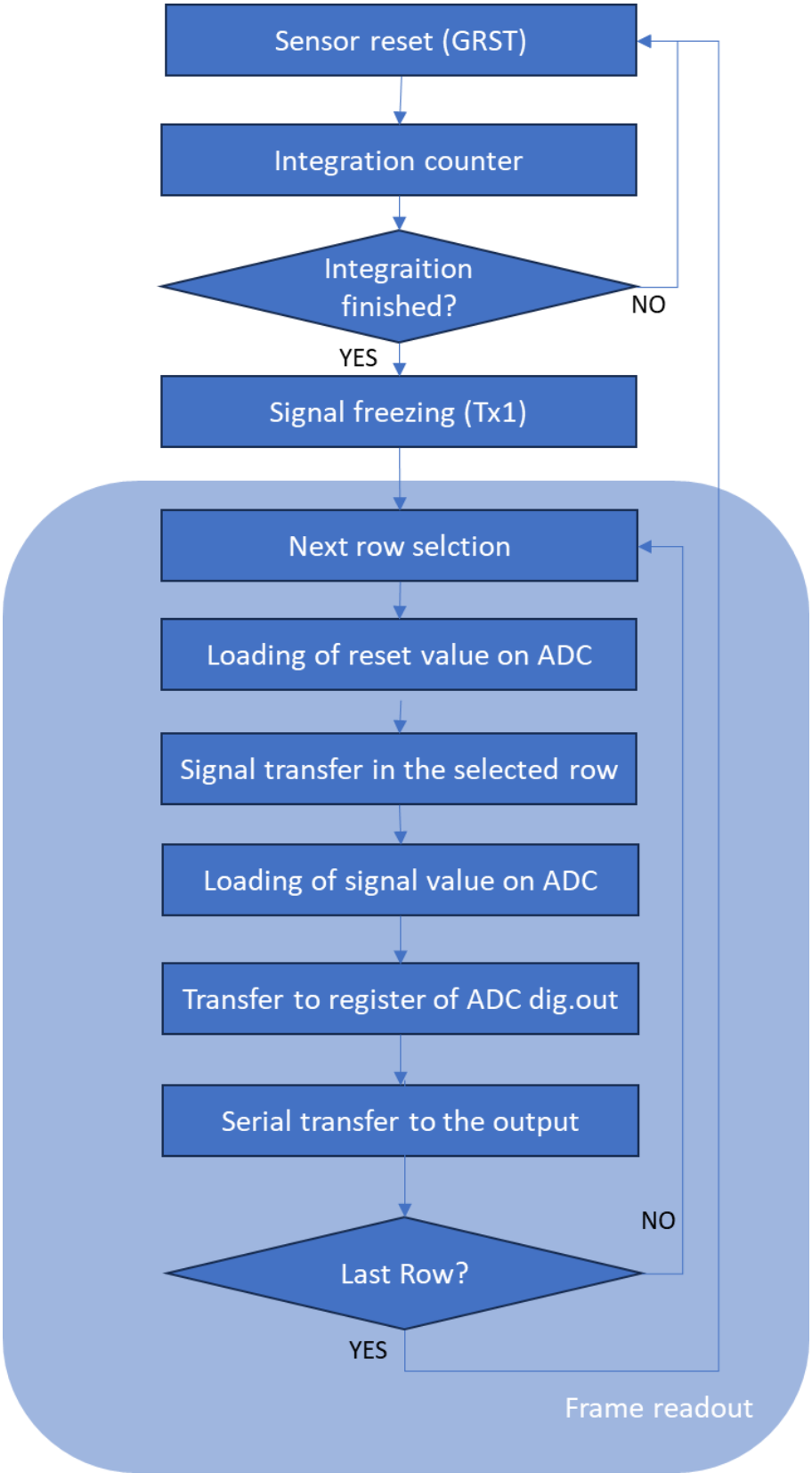
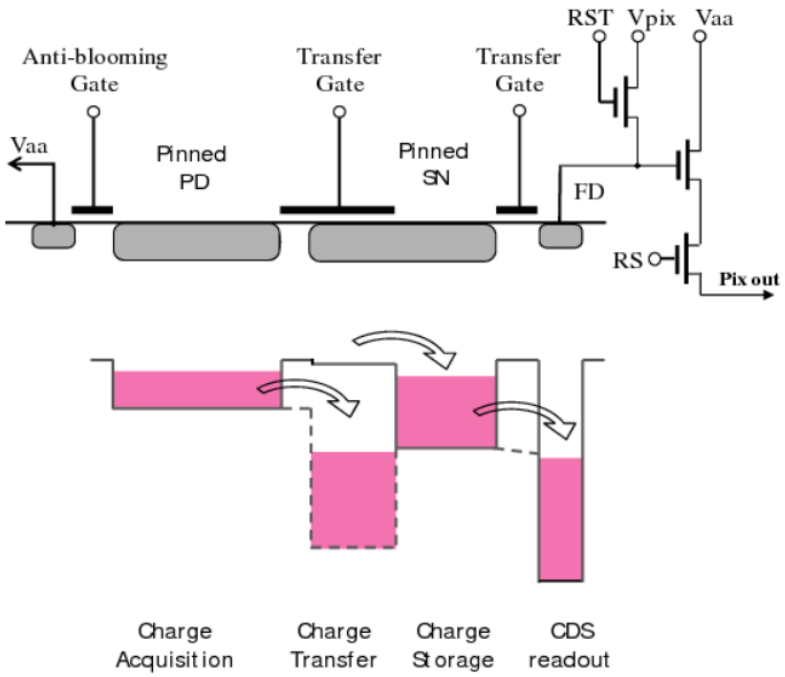


Fig. 5.3 Flow diagram describing the operations of the sensor.

5.1.1. Pixel Array



Supply	Level
Vdd Pix	3V
GRST Hi	3.3V
GRST Low	0V
RST Hi	3.5V
RST Low	0V
SEL Hi	3.5V
SEL Low	0V
TX1 Hi	3.5V
TX1 Low	-0.5V
TX2 Hi	3.3V
TX2 Low	-0.5V

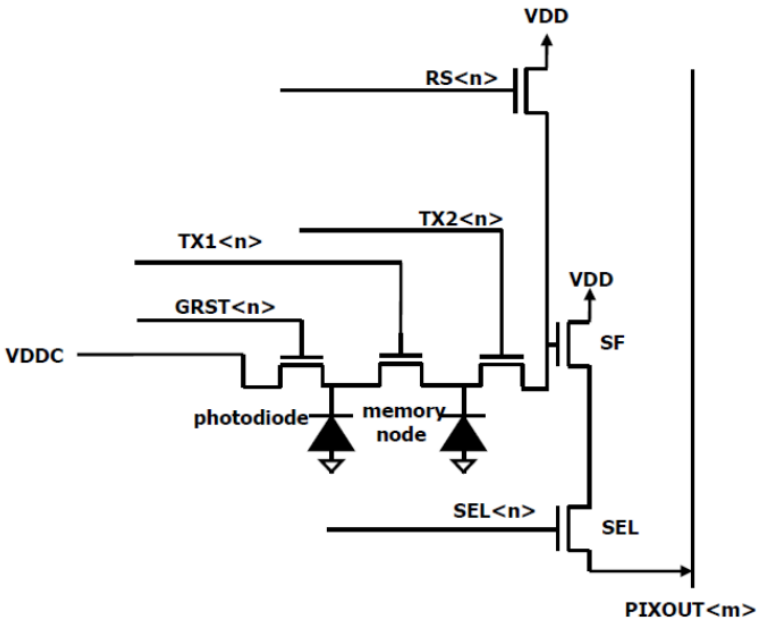


Fig. 5.4 Pixel configuration.

The array of pixels is made by 800x600 square pixels with a pitch of 4.8um. Each pixel contains a photodiode that converts the incoming light into a continuous voltage. The pixel architecture is a standard pinned photodiode with flush/shutter gate and global shutter (3-transistor, 1 transfer gate, 1 shutter gate and 1 photodiode, Fig. 5.4). High dynamic range (HDR) is achieved by conditioning the shutter gate with a step-wise signal, applied to the GRST terminal, externally generated.

The pixel configuration guarantees a programmable dynamic range from 65 dB to over 100 dB by changing the shape of the conditioning signal VRES.

Tab. 5.1 Pixel characteristics

Pixel type	Global shutter 6T CDS
Technology	CIS 180nm FSI
Conversion gain	110 uV/e-
Full-well	8.5 ke-
Native dynamic range	65dB
Noise	4e-

5.1.1.1. Conditioning signal GRST

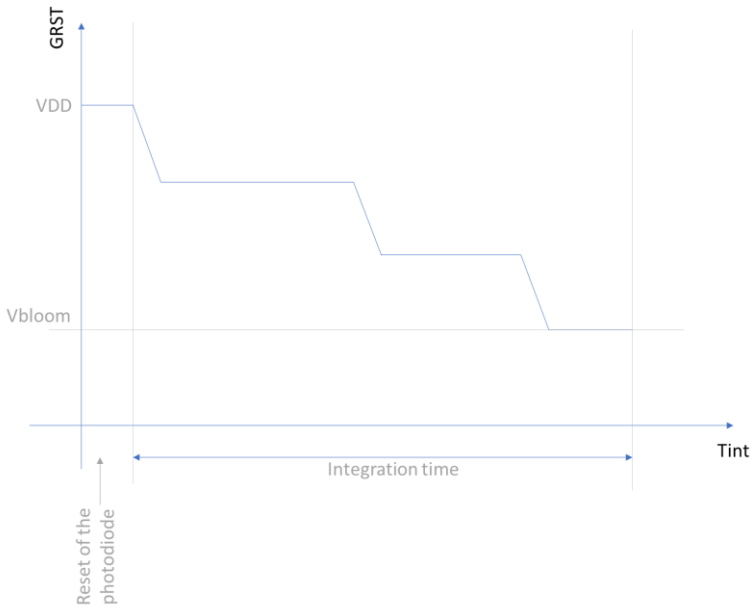


Fig. 5.5 GRST ramp generation shape.

GRST is a special analog reference voltage for the pixel array. GRST is changed during the integration time according to a piece-wise-linear non-increasing curve. At the end of the integration time the photodiode signal is freezed by a transfer to the floating gate (FD) and ready for readout. Because of physical decoupiling of the photodiode (PD) junction from the readout of the pixel, another integration and another conditioning curve can start in parallel to readout.

The GRST is generated externally, and a possible shape is shown in Fig. 5.5. The only constraint for the ramp generator is that the GRST value cannot be increasing, therefore there can be plateaus.

The integration time can be between 1msec and 100msec with a typical value of 30msec.

5.1.2. Video Line

When a row-decoder selects a row, every pixel of the selected row transfers simultaneously its output value to the corresponding column ADC which stores the reset value. Then each pixels of the selected row transfer its signal from FD to the output and their signal values are transferred to the corresponding ADC. In this way the two samples are correlated. The ADC subtract the reset value from the stored signal and digitalize the result on a 10-bit word, available at the output. Such subtraction is a standard technique to mitigate *Fixed Pattern Noise (FPN)* (see section 5.1.3 for more details). Since the correlation between the two samples, also Pixel temporal noise is subtracted.

The output of the ADCs are transferred to a parallel register, sequentially selected for data transfer through the 4-line MIPI interface.

5.1.3. Fixed Pattern Noise

Fixed Pattern Noise (FPN) is a time-invariant noise in images. The FPN is caused by the non-uniformity of the transistor's characteristics within repetitive devices like pixels and column amplifiers, resulting from fabrication process tolerances. Threshold voltage variations introduce a voltage-offset characteristic of each pixel and gain variations introduce a gain error. As described above, pixel FPN offset subtraction is performed inside the sensor by CDS blocks while column FPN generated by the CDSs must be subtracted externally by double sampling the signals, first during the signal phase and after during the reset phase. However, it is recommended to perform also a subtraction using software-based techniques.

5.1.4. Decoders

Decoder are used to address the pixels of the array. Five decoders are used inside the sensor:

- one for row selection (RowSEL);
- one for row reset (RowRES);

- one for row signal transfer (RowTx2)

The decoders are based on a shift register architecture with access to the data I/O of the first and last cells only. Each decoder can be synchronously set to 0 (by the SEL_S0 signal and decoders enable, SEL_EN), while the output signal going to the pixel array can be asynchronously forced to 0 by the SEL_FT0 signal without affecting the logical value stored inside. The decoder acts on the rising edge of the decoders signal enable, SEL. A schematic view of a decoder cell is shown in Fig. 5.6.

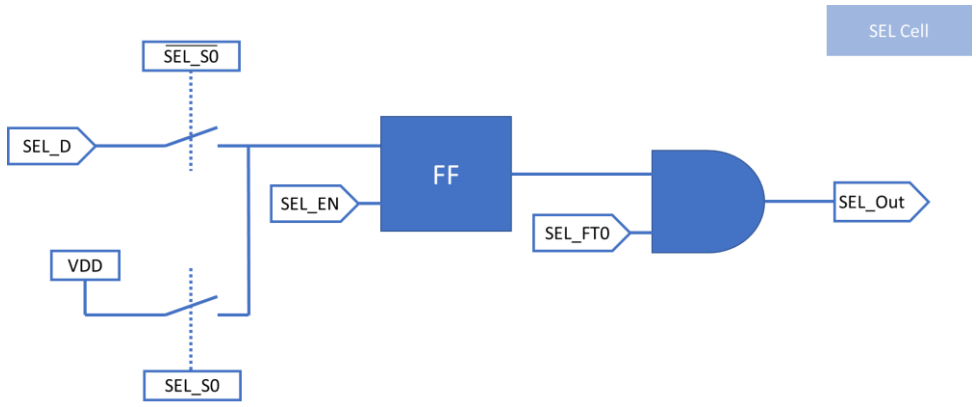


Fig. 5.6 Schematic diagram of a decoder cell

5.1.5. Column ADC

5.1.6. Analog references

The needed analogue references are:

- 4 current references: IBiasCDS, IBiasCOL, IColADC
- 3 voltage references: ColADC_VREF, GRST, Vpix.
- 2 over/under voltages: VDDH, VSSH.

All the reference are internally generated starting from a stable Band Gap reference or from the supply.

Typical values are reported in the following tables

Tab. 5.2 Typical values for current (above) and voltage (below) references.

	Description	Typ.Value (μA)	Load
IBiasCOL	Bias current for bit-lines	5	200Ohm
IBiasCDS	Bias current for column CDSs	5	200Ohm
IColADC	Bias current for Column ADC	?	?

	Description	Typ. Value	Load
ColADC_VREF	Bias voltage for column ADC	?	?
GRST	Global blooming and dynamic control	Dynamically Programmable Range: 3.3V-0V[1]	Low current load (minimum gates)
Vpix	Pixel Array power supply	3.0V	High current load
VSSH	Negative charge tranfer reference voltage	-0.5V	External load to be added to make it stable
VDDH	Positive charge tranfer reference voltage	3.5V	External load to be added to make it stable

The 3 currents will be generated mirroring the current from a programmable resistance (3-bit).

GRST will be generated by a Ladder DAC with output buffer. The buffer will have to supply a low current, high capacitive load.

Vpix will be generated starting from the Band Gap reference voltage

6. EYE-TECH Digital core

The digital core of **ET2-GS-FC** is responsible for the full operability of the chip and includes the controller, the I2C-like input interface, the output data serializer and the MIPI output interface. Further to this, PLL for clock generation is included.

6.1. Operating modes

There are three possible operating modes:

1. **Normal Operation Mode**: the imager is running, all digital and analogue modules are working. The chip enters this mode when the user writes the "start" instruction by the I2C-like interface;
2. **Stand-By Mode**: the chip acquisition/read-out is stopped. The analogue core continues running. No clock gating is foreseen. The imager enters this mode when the user writes the "stop" instruction via the I2C-like interface.

[1] Refer to section 5.1.1.1 for more details about the curve generation.

3. **Power-Down Mode**: the imager is stopped and all internal registers are brought back to the idle state. The analogue core is powered down. The chip presents the lowest power consumption. The imager I2C-like interface does not work. The chip enters this mode on the rising edge of the external asynchronous reset signal (RSTN). The imager exits this mode on the falling edge of the same signal.

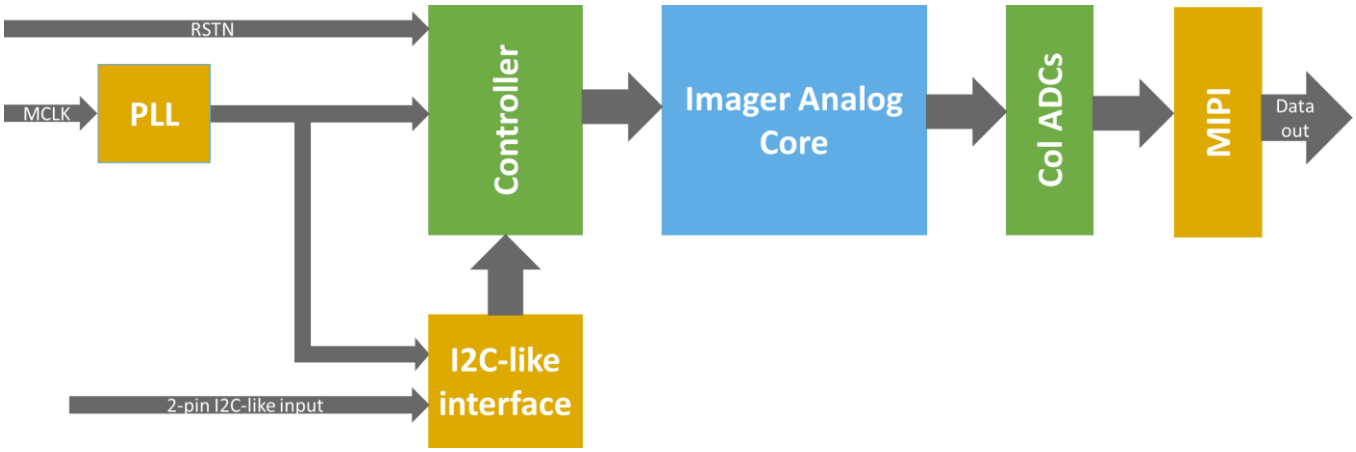


Fig. 6.1 EYE-TECH imager digital core block diagram.

The block diagram of the Digital Core is shown in Fig.6.1. It is made by two main blocks:

1. **PLL**: generated the internal high frequency master clock starting from external low frequency reference;
2. **I2C-like input interface**: manages the communication between the imager and the external HOST. The imager is configured as I2C SLAVE.;
3. **Internal controller**: this module handles the signals controlling the array and the selection/reset decoders. Moreover, it drives the configuration register, as well as the trimming of internal analog reference generation.
4. **MIPI**: this block is responsible for the output signal management with standard protocol.

6.2. I2C

The digital communication interface between the ET2-GS-FC imager and the external control unit (HOST) is based on an I2C-like protocol, which allows to write/read logic internal register. The communication interface is based on two signals: *scl*[2] (input) and *sda* (input/output).

The whole architecture works at the rising edge of the master clock.

6.2.1. Communication Protocol

The I2C-like serial communication protocol is shown in Fig. 5.10. The imager is always the slave of the I2C-like protocol while the master (the HOST) can write/read the registers.

[2] Note that the scl signal can run up to 800 kHz in frequency.

A write operation is based on the following steps:

- START state (*scl* signal high combined with a transition from high to low of the *sda* line);
- slave address transmission (8 bits);
- wait for ACKN (negative acknowledge, 1 bit);
- transmission of the register address (8 bits);
- wait for ACKN;
- data register transmission (8 bits);
- wait for ACKN;
- STOP state (transition of the *sda* line from low to high while *scl* is high).

A read operation is based on the following steps:

- START state (*scl* signal high combined with a transition from high to low of the *sda* line);
- slave address transmission (8 bits);
- wait for ACKN (negative acknowledge, 1 bit);
- address register transmission (8 bit)
- wait for ACKN
- REPEATED START state (SCL signal high combined with a transition from high to low of the SDA signal);
- slave address transition hidden through 0x01(8 bit);
- wait for ACKN
- wait for address register transmission (8 bits);
- ACKN transmission;
- wait for data register (8 bits);
- ACKN transmission;
- STOP state (transition of the *sda* line from low to high while *scl* is high).

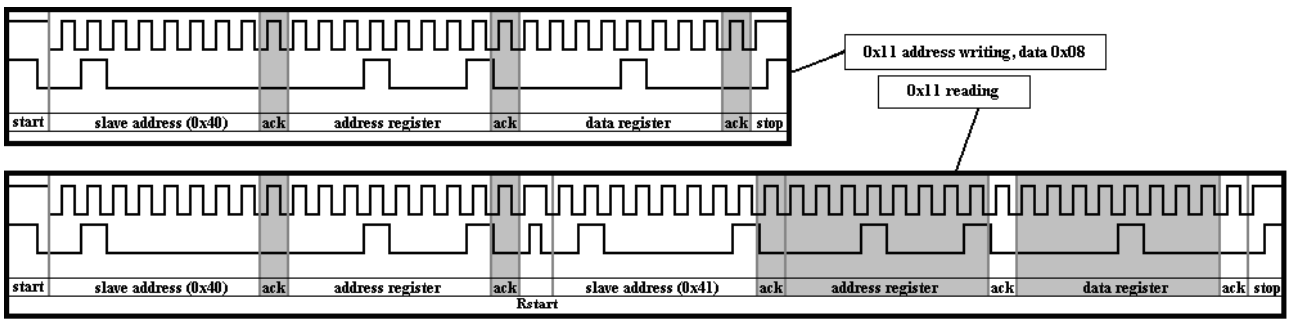


Fig. 6.2 Temporal sequence of I2C read and write operations.

6.2.2. Register Map

Tab. 6.1 EYE-TECH register map

I2C addr	Acronym	Possible values	R/W[3]	Description	Default
0x86	Instr_ReadMode	0x01/0x04 /0x10/0x40	R/W	Read-out Mode: 0x01 = Normal Sampling – 0x04 = Sub-Sampling 4 0x10 = Sub-Sampling 16 – 0x40 = Sub-Sampling 64	0x01
0x87	Instr_PowerDown	0x01/0x02	R/W	Set the imager power down state: 0x01 = Normal Operation (Configuration register 0xF00008C6) 0x02 = Power down State (Configuration Register 0x00008000)	0x01
0x88[3]	Instr_SetReg1	0xXX	RW	Set the 1st byte of the Configuration register: Configuration register (7:0)	0xC6
0x89[3]	Instr_SetReg2	0xXX	RW	Set the 2nd byte of the Configuration register: Configuration register (15:8)	0x08
0x8A[3]	Instr_SetReg3	0xXX	RW	Set the 3dr byte of the Configuration register: Configuration register (23:16)	0x00
0x8B[4]	Instr_SetReg4	0xXX	RW	Set the 4th byte of the Configuration register: Configuration register (31:24)	0xF0
0x8D	Instr_SetInteg1	0xXX	R/W	Most significant byte of the 16-bit Integration Time Register	0x05
0x8E	Instr_SetInteg2	0xXX	R/W	Least significant byte of the 16-bit Integration Time Register	0xB9
0x8F[5]	Instr_Start_Col1	0xXX	R/W	Most Significant byte of the 16-bit register defining the first column address for the read-out.	0x00
0x90[4]	Instr_Start_Col2	0xXX	R/W	Least Significant byte of the 16-bit register defining the first column address for the read-out.	0x00
0x91	Instr_End_Col1	0xXX	R/W	Most Significant byte of the 16-bit register defining the last column address for the read-out.	0x01

[3] "R/W" means that the register can be wrote and read; "RO" means "read only".

[4] these instructions have to be sent sequentially; since all the four registers are set in sequence, the refresh of the Configuration Register is activated. See section 5.10.1.6 for possible values.

[5] It is mandatory that the starting addresses for row and column were even in order to guarantee the mosaic integrity. This check is mandatory for the host side as no automatic control was embedded.

0x92	Instr_End_Col2	0xXX	R/W	Least Significant byte of the 16-bit register defining the last column address for the read-out.	0xFF
0x93[4]	Instr_Start_Row1	0xXX	R/W	Most Significant byte of the 16-bit register defining the first row address for the read-out.	0x00
0x94[4]	Instr_StartRow2	0xXX	R/W	Least Significant byte of the 16-bit register defining the first row address for the read-out.	0x00
0x95	Instr_EndRow1	0xXX	R/W	Most Significant byte of the 16-bit register defining the last row address for the read-out.	0x01
0x96	Instr_EndRow2	0xXX	R/W	Least Significant byte of the 16-bit register defining the last row address for the read-out.	0xFF
0x99	Instr_SetRegRED	0xXX	R/W	Gain of the DDS for the RED channel	0x20
0x9A	Instr_SetRegGREEN1	0xXX	R/W	Gain of the DDS for the GREEN1 channel	0x20
0x9B	Instr_SetRegGREEN2	0xXX	R/W	Gain of the DDS for the GREEN2 channel	0x20
0x9C	Instr_SetRegBLUE	0xXX	R/W	Gain of the DDS for the BLUE channel	0x20
0x9E	Instr_VREF	0xXX	R/W	VREF DAC	0x78
0xA1	Instr_VCM	0xXX	R/W	VCM DAC	0x78
0xA2	Instr_VBCDS	0xXX	R/W	VBCDS DAC	0x96
0xA3	Instr_IBIASCDS	0xXX	R/W	IBIASCDS DAC	0x00
0xA4	Instr_IBIASCOL	0xXX	R/W	IBIASCOL DAC	0xBB
0xA5	Instr_IBVAMP	0xXX	R/W	IBVAMP DAC	0xBB
0xA6	Instr_IBOUTAMP	0xXX	R/W	IBOUTAMP DAC	0x00
0xAA	Instr_IBIAS_DAC1	0xXX	R/W	IBIAS_DAC1 DAC	0xBB
0xAB	Instr_IBIAS_DAC2	0xXX	R/W	IBIAS_DAC2 DAC	0x00
0xB1	Instr_VREF2	0xXX	R/W	VREF2 DAC	0x78
0xB2	Instr_VCM2	0xXX	R/W	VCM2 DAC	0x78
0xB7	Instr_GRST_Vini	0xXX	R/W	Initial value for the VRES ramp generation	0xFF
0xB8	Instr_ GRST _V0	0xXX	R/W	1st voltage reference for the VRES ramp generation, V_0	0xD0
0xB9	Instr_ GRST _V1	0xXX	R/W	2nd voltage reference for the VRES ramp generation, V_1	0xA0
0xBA	Instr_ GRST _V2	0xXX	R/W	3rd voltage reference for the VRES ramp generation, V_2	0x8D
0xBB	Instr_ GRST _Vf	0xXX	R/W	Final voltage for the VRES ramp generation, V_f	0x72
0xBC	Instr_ GRST _T0	0xXX	R/W	1st time reference point for the VRES ramp generation, T_0	0x54
0xBD	Instr_ GRST _T1	0xXX	R/W	2nd time reference point for the VRES ramp generation, T_1	0xA9

0xBE	Instr_ GRST _T2	0xXX	R/W	3rd time reference point for the VRES ramp generation, T_2	0xCB
0xC5	Instr_GetID	-	RO	Get the firmware version ID	0x01
0xC8	Instr_EnableSignal	0xXX	R/W	Enable/disable Pull_up_bus (Bit 0=1 enabled) and Pull_Down_Col (Bit 1=1 enabled)	0x03
0xD0	Instr_SReg	0xXX	RO	Read the status register	-

6.2.3. Controller

The *Controller* is the core logic module which drives the analogue core of the imager (Fig. 6.3).

It presents a hierarchical architecture which is based on different sub-modules that drive specific analogue parts of the imager. In the following paragraphs, a complete description of the main sub-modules, as well as the logic top-level, are described.

6.2.4. Top-Level Architecture

The *Controller* architecture works on the rising edge of the master clock. The module presents an external active-high reset signal called *RSTN*, which allows a hardware reset of the whole architecture.

The *Controller* is connected to both external pads (*EX_**) used for a manual configuration of the core and internal blocks used for the automatic smart control. It presents five internal sub-modules which interface with the Controller main process: Ana_Ref, RegConfig_int, RegCol_int, Chip_int and VRES_int.

The *Controller* top-level process maps all internal registers that are addressable via the I2C-like interface (see Tab. 6.2 for more details).

The main function of the *Controller* internal process is to control the state of the whole digital architecture (see section 6.1 for possible operating modes).

When **the *rstn* input is high** the digital core is reset and the analogue core is pulled into the power down state.

Once **the *rstn* signal is pulled low**, the digital core exits the reset state and the I2C-like communication interface is enabled. At this point the user has to issue the *Intr_PowerDown* command in order to let the analogue core exit the power down state.

The digital core enters in the normal operation state only after the *Instr_SetAcq* register is set to 0x02.

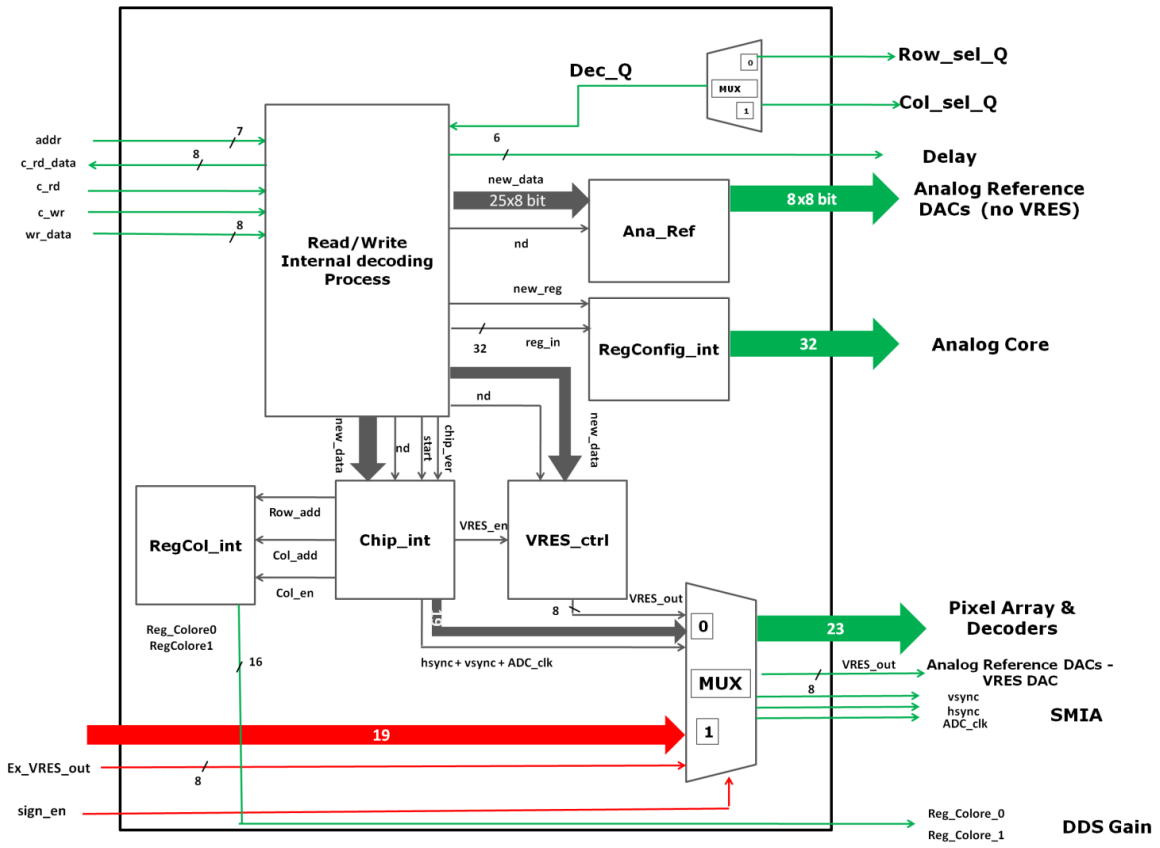


Fig. 6.3 EYE-TECH_Controller top-level architecture. TBU

If the user writes 0x01 the *Controller* drives low the internal start signal and the digital core is in a **software reset** state: the alalog reference trimming and control still remain active while the GRST generation and the digital control signals are stopped and reset.

It is worth mentioning that the automatic control signal generation by the controller can be bypassed by setting the pin *sign_en* to logic level high. Indeed, when sign_en is set high, the sub-blocks *Chip_int* and *VRES_ctrl* are bypassed (see the block diagram in Fig. 5.8).

6.2.5. Integration time setting

The user can decide the integration time of the imager by setting the value of the register *Instr_SetInteg1* and *Instr_SetInteg2* (see Tab. 5.4).

This value is translated internally into a 16-bit unsigned integer; and multiplied by 2048, thus allowing to obtain a 27-bit counter (Fig. 6.4) which is suitable for representing the required integration time range.

b 7	b 6	b 5	b 4	b 3	b 2	b 1	b 0	b 7	b 6	b 5	b 4	b 3	b 2	b 1	b 0	0	0	0	0	0	0	0	0	0	0	0
Instr_SetInteg1								Instr_SetInteg1								0										

Fig. 6.4 27-bit integration time counter

The value of *Instr_SetInteg1* and *Instr_SetInteg2* registers has to be calculated as follow:

$$time_integ = \frac{integration_time(s)}{mclk_period(s) * 2048}$$

Where *time_integ* is the final value to divide between the two registers and *mclk_period* is the system master clock in seconds.

It is worth mentioning that the final frame rate is is the higher between the integration and the readout time. Indeed, the maximum frame rate is limited at 45 fps, that is the maximum achievable when the limit is the readout time.

The readout period corresponds to the high period of the VSYNC signal. For the maximum resolution of 1920x1080 it is expected to be 22.2msec.

6.2.6. Sub-windowing and sub-sampling modality

Important features of the ET2-GS-FC are sub-windowing and sub-sampling modalities.

For sub-windowing the user can select both the starting and ending row/column by setting the proper registers via the I2C-like interface (see Tab. 6.2). The only constraint for the user is that the starting and ending row must be even numbers in order not to alter the mosaic integrity.

Sub-sampling allows to reduce the imager resolution of one-fourth, one-sixteenth and one-sixty-fourth. The Controller jumps quickly the undesired rows/columns by driving the selection decoders and reads the other ones till it reaches the user-defined final row/column. Fig. 6.5 shows the algorithm implemented.

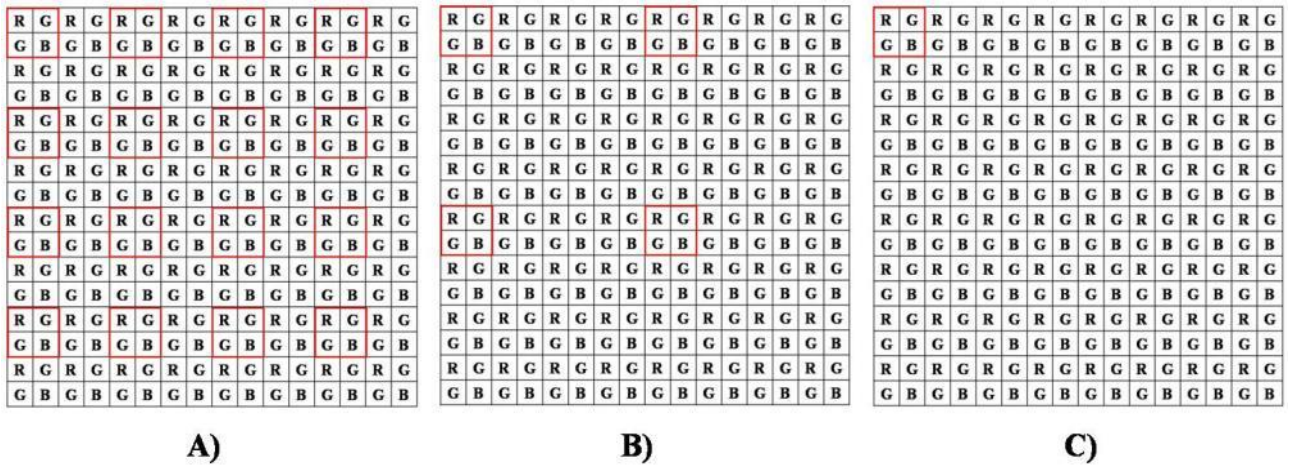


Fig. 6.5 The sub-sampling algorithm for the ET2-GS-FC chip to provide the host one fort (A), one sixteenth (B) or one sixty-fourth (C) of the whole chip array. Red-edged pixels are selected while the other are skipped.

The register *read_mode* sets the read-out mode of the *Chip_int* FSM. Four possible read mode have been implemented (Tab. 5.6). It is mandatory that this register shows one of these values, otherwise the internal controller fails.

Tab. 6.2 Read-out Modalities.

Read_mode value	Read-out Mode
0x00	Normal Sampling
0x04	Subsampling4: each 4 columns (rows) only the first two are read while the other are skipped (configuration A in Fig. 5.13).
0x10	Subsampling16: each 8 columns (rows) only the first two are read while the other are skipped (configuration B in Fig. 5.13).
0x40	Subsampling64: each 16 columns (rows) only the first two are read while the other are skipped (configuration C in Fig. 5.13).

6.2.6.1. Chip_int sub-module

Chip_int is a sub-block of the *Controller* module and it generates the control signals for the analog core and the output data synchronization signals and drives the RegCol_int and GRST_ctrl submodules. The clock is provided by the PLL.

The *Chip_int* sub-module presents a pool of internal registers, that are summarized in Tab. 6.4. An internal process manages the update of these registers when a new I2C-command is received and it refreshes their values every 65000 master clock cycles.

Tab. 6.3 Chip_int internal register list.

Register Name	I2C address	Register width	Upgraded During	Register Function
Instr_SetInteg1/2	0x8D 0x8E	16	read-out (led_out = 0, vsync = 1)	User-defined 16-bit integration time (this register is translated internally into an unsigned integer)
Instr_Start_Row1/2[6]	0x93 0x94	16	integration (led_out = 1, vsync = 0)	User-defined 16-bit starting row address (this register is translated internally into an unsigned integer). This register is used to enable sub-windowing.
Instr_EndRow1/2	0x95 0x96	16	integration (led_out = 1, vsync = 0)	User-defined 16-bit final row address (this register is translated internally into an unsigned integer). This register is used to enable sub-windowing.
Instr_Start_Col1/2	0x8F 0x90	16	integration (led_out = 1, vsync = 0)	User-defined 16-bit starting column address (this register is translated internally into an unsigned integer).

[6] It is mandatory that the 2-bytes integer has to be even in order not to break the mosaic integrity

				This register is used to enable sub-windowing.
Instr_End_Col1/2	0x91 0x92	16	integration (led_out = 1, vsync = 0)	User-defined 16-bit final column address (this register is translated internally into an unsigned integer). This register is used to enable sub-windowing.
Instr_ReadMode	0x86	8	integration (led_out = 1, vsync = 0)	This register enable the user to set the read-out mode: either normal sampling or sub-sampling (4, 16, 64)

6.2.6.2. GRST_ctrl sub-module

The GRST *register* is used to store the **GRST voltage**, which is a special analogue reference voltage for the pixel array. The reset voltage is generated along the integration time varying as a non-ascending ramp. If the integration time is lower than the readout time, the GRST value at the beginning of the integration is held high to compensate the difference to the readout time. As a consequence, the end of integration and readout will match. If the integration time is longer than the readout time, readout will be placed in stand by till the end of the integration. At the end of the integration time, its final value *vf,* is globally transferred with a pulse of Tx1 (see xxx for more details) and readout can stat again.

A possible shape of the GRST_out signal is shown in Fig. 5.14. The only constraint for the ramp generator is that the GRST_out value has not to be crescent, therefore there can be plateaus if defined by the host controller. The user can define each parameter shown in Fig. 5.14: the initial and the final values, the three defined (both in voltage and time) points in which the ramp must pass through and the slope of the ramp in each interval.

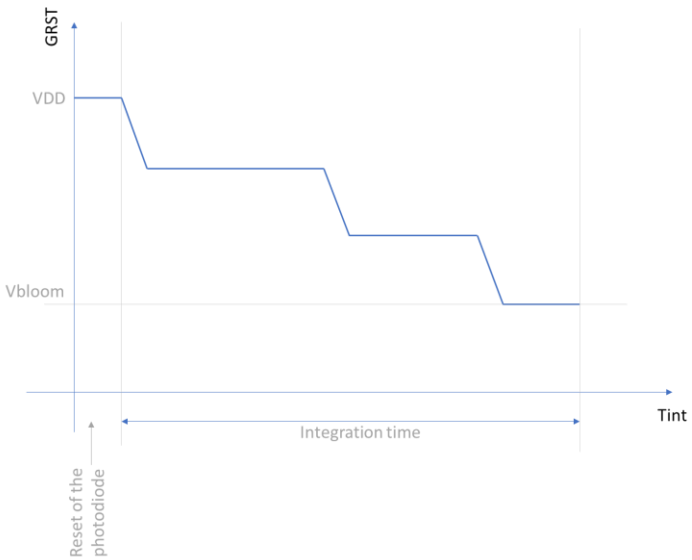


Fig. 6.6 VRES ramp generation shape.

Note that the x-axis of the graph is not an absolute time scale. Indeed, the time is divided into 256 steps in order to comply with the length of the time reference register.

$$Step_Length[mclk_cycles] = \frac{t_{int(16-bit)} \times 2048}{256}$$

$$Step_Length[s] = \frac{t_{int(16-bit)} \times 2048}{256} \times T_{mclk}[s]$$

It is worth mentioning that time reference values should require proportions for their calculation, thus requiring real numbers. Therefore, their calculation is demanded to the host controller of the chip, while the EYE-TECH1080 imager accepts them as 8-bit integer values. All internal registers of the VRES_ctrl sub-module are summarized in Tab. 5.8.

Tab. 6.4 VRES_ctrl internal register list.

Register name	Register width (bit)	Default value	Funcion
Vini	8	0xFF	VRES ramp starting value
v0	8	0xD0	VRES ramp first voltage reference point
t0	8	0x54	VRES ramp first time reference point
dv0	8	0x01	VRES ramp first interval slope
v1	8	0xA0	VRES ramp second voltage reference point
t1	8	0xA9	VRES ramp second time reference point
dv2	8	0x01	VRES ramp second interval slope
v2	8	0x8D	VRES ramp third voltage reference point
t2	8	0xCB	VRES ramp third time reference point
dv2	8	0x01	VRES ramp third interval slope
Vf	8	0x72	VRES ramp end voltage
dv3	8	0x01	VRES ramp forth interval slope
t_integ	16	0x05B9	16 bit integration time

All these internal registers are upgraded only during the imager read-out phase, when the ramp generation is blocked. This behaviour is transparent for the host controller. An internal update process refreshes the values of each internal register every 65000 master clock cycles.

6.2.6.3. RegConfig_Int sub-module

This block is responsible for controlling the Configuration Register, a 32-bit register having test, control and power-down functionalities.

The initialization value of the Configuration Register is set to Powerdown thus putting the imager in a power down state when the reset input is high.

A list of the Configuration Register pins is reported in Tab. 6.5.

Tab. 6.6 Configuration register map

Configuration Register Pin number	Name	Function	Normal operation	Power down mode
0	DDS_TST1	Reserved	0	0
1	DDS_TST2	Reserved	1	0
2	DDS_TST3	Reserved	1	0
3	DDS_TST4	Reserved	0	0
4	DDS_TST5	Reserved	0	0
5	DDS_TST6	Reserved	0	0
6	DDS_TST7	Reserved	1	0
7	DDS_TST8	Reserved	1	0
8	DDS_TST9	Reserved	0	0
9	DDS_TST10	Reserved	0	0
10	DDS_TST11	Reserved	0	0
11	TST-CDS_EN	Reserved	1	1
12	TST-CDS_ENN	Reserved	0	0
13	Q_SEL0	Reserved	0	0
14	Q_SEL1	ANA_TST = VRES	0	0
15	RF_TST1	ANA_TST = VCM	0	0
16	RF_TST2	ANA_TST = VREF	0	0
17	RF_TST3	ANA_TST = VbCDS	0	0
18	RF_TST4	ANA_TST = IBIASCOL	0	0
19	RF_TST5	ANA_TST = IBIASCDS	0	0
20	RF_TST6	ANA_TST = IBIAS_DAC	0	0
21	RF_TST7	ANA_TST = IBOUTAMP	0	0
22	RF_TST8	IBVAMP	0	0
23	RF_TST9	Reserved	0	0
24	RF_TST10	Reserved	0	0
25	RF_TST11	Reserved	0	0
26	RF_TST12	Reserved	0	0
27	RF_TST13	Reserved	0	0
28	PWD1	Active-low power-down	1	0
29	PWD2	Active-low power-down	1	0
30	PWD3	Active-low power-down	1	0
31	PWD4	Active-low power-down	1	0

Tab. 6.5 Configuration Register pin list

Note that there are four power-down signals, thus allowing to turn off selectively different blocks of the imager.

Four registers are used to configure this internal configuration register (Tab. 5.10).

Register Name	I2C addr	Register width	Upgraded During	Register Function
Instr_SetReg 1	0x88	8	When the all the 4 commands are sent	Configuration register (7:0)
Instr_SetReg 2	0x89	8	When the all the 4 commands are sent	Configration register (15:8)
Instr_SetReg 3	0x8A	8	When the all the 4 commands are sent	Configration register (23:16)
Instr_SetReg 4	0x8B	8	When the all the 4 commands are sent	Configration register (31:24)

The configuration register is updated only when the last byte is written, i.e. when the user has sent all the four command configuring the configuration register via the I2C-like interface. This behaviour is implemented in order not to drive the analogue core with spurious or conflicting configurations.

The multiplexer control signal Dec_Q enable the test output of the selection decoders and can be configured by writing into the configuration register the values reported in Tab. 6.7, thus enabling on the output Dec_Q the output signature of the column or row decoders (Fig. 6.3).

Tab. 6.7 Dec_Q truth table

Dec_Q value	Q_SEL0 - Q_SEL1
Row_sel_Q	1 0
Col_sel_Q	0 1
0	Otherwise

6.2.6.4. Ana_Ref sub-module

This block drives the DAC Registers of the imager. There are 8 registers which drive the corresponding DACs to provide a complete set of currents and voltages reference values. The Analog reference internal registers are summarized in Tab. 5.12.

Tab. 6.8 DAC driving registers.

Register Name	Function	Default value in Normal Operation
IBIAS_DAC1	DAC bias current	187
IBIASCOL	Column Bias current	0
IBIASCDS	CDS Bias current	150
IBVAMP	Amplifier Bias current	187
IBOUTAMP	Output Amplifier Bias Current	187
VCM	Amplifier common mode voltage	120
VREF	Pixel Reference Voltage	120
VBCDS	Column CDS Bias voltage	15

All these registers are stored in both the Controller top-level and in the Ana_Ref sub-module (where they are connected to the corresponding DACs). The DAC registers are refreshed with a refresh rate of 65000 master clock cycles.

6.2.6.5. RegCol_int submodule

This sub-module aims to drive the colour gain DAC block which is embedded into the DDS. Four different gain values can be set and are changed dynamically during the readout: Red (R), Green1 (G1), Green2 (G2) and Blue (B). Since the colour filters which are placed upon the pixel array present a fixed patter, this gain value is strictly related to the position of the pixel into the array, as shown in Fig.5.15.

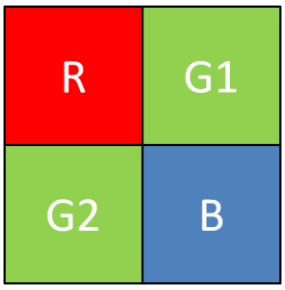


Fig.6.7 Color Filters Pattern.

Therefore, an internal register is dedicated to each colour and can be adjusted by the user as reported in Tab. 5.1. The RegCol_int internal registers are summarized in Tab. 5.13.

Tab. 6.9 RegCol_int sub-module internal register

Register name	Register number	Register width (bit)	Default value	Funcion
RegRED	0x99	8	0x20	Red channel gain
RegGREEN 1	0x9A	8	0x20	Green1 channel gain
RegGREEN 2	0x9B	8	0x20	Green2 channel gain
RegBLUE	0x9C	8	0x20	Blue channel gain

6.2.6.6. Mux sub-module

This block is an asynchronous multiplexer. Depending on the value of the external control signal *sig_en* all control signals to the pixel array, the GRST DAC, the decoders, and the Colour Gain come from the external pads (sig_en = 1) or from the Controller sub-modules (sig_en = 0).

Please contact EYE2DRIVE support service for more details on the manual configuration.

7. Column ADC

8. Output interface – MIPI CSI-2 Tx D-PHY

The MIPI CSI-2 transmitter is composed of two major subsystems:

- CSI-2 Controller (Digital logic):
- D-PHY TX (Mixed-Signal / Analog Front-End)

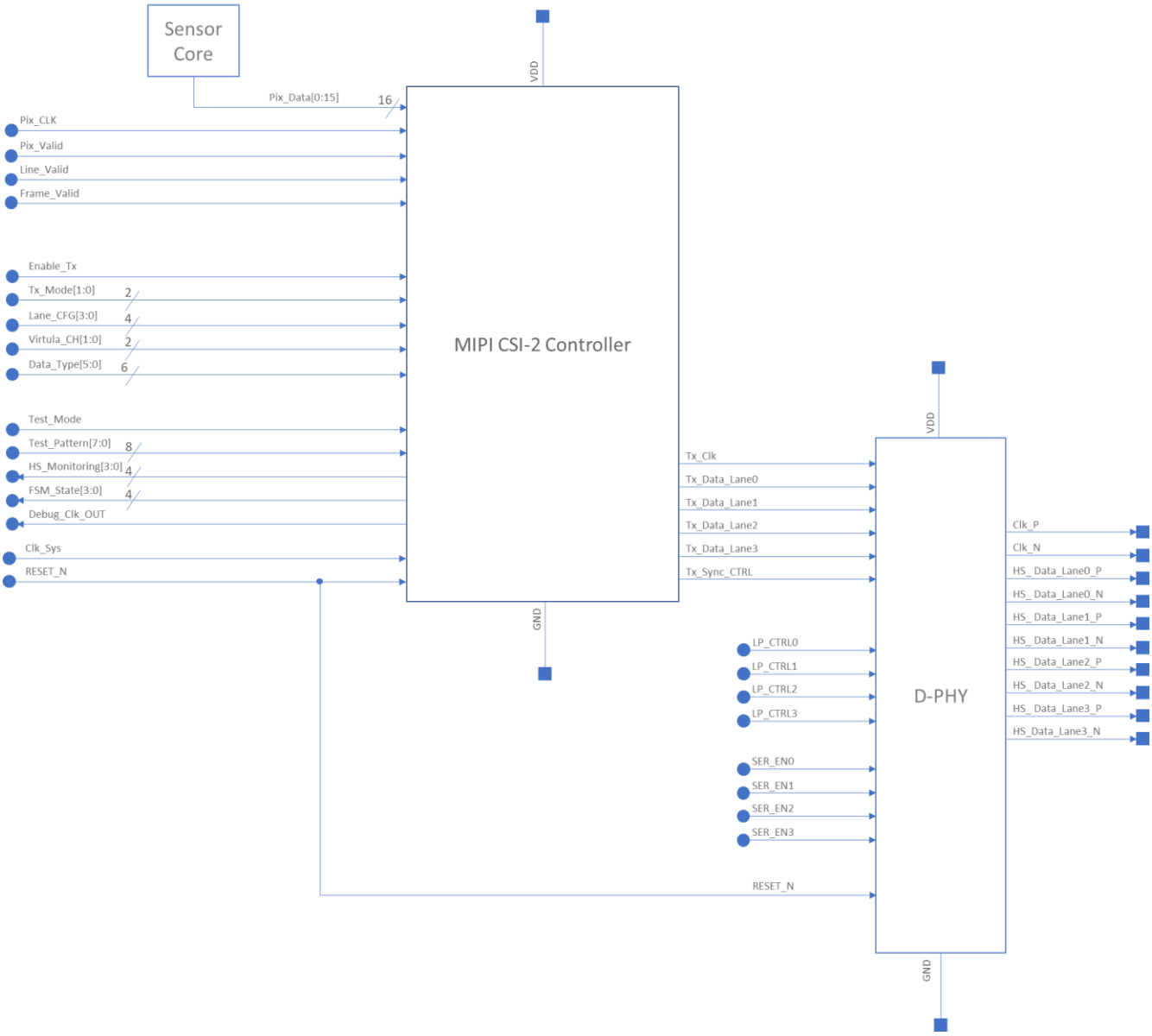


Fig. 8.1 Block diagram of the MIPI interface<

MIPI CSI-2 PHY Tx architecture is depicted in.

The CSI-2 controller is 100% digital and can be implemented in RTL (VHDL/Verilog).

The PHY output requires analog design, mainly for the differential high-speed driver (LVDS-like) and LP mode support.

A custom differential analog driver for the HS signals must be implemented.

Compliance with MIPI D-PHY requires attention to signal swing, common-mode, LP/HS transitions, and ESD at the pad level (refer ESD document from Tower).

Tab. 8.1 Complete interface signal MAP

Signal Name	Direction	Int.	Description
Column-ADC Interface			
`Pix_Clk`	I-D		Pixel clock used to latch `Pix_Data`.
`Pix_Data[0:15]`	I-D	ADC	Parallel pixel data bus.
`Pix_Valid`	I-D	FSM	Indicates valid pixel data on `Pix_Data`.
`Line_Valid`	I-D	FSM	Indicates active video line
`Frame_Valid`	I-D	FSM	Indicates active video frame
Control and Configuration			
`CLK_SYS`	I-D	FSM	System clock for internal logic and configuration registers.
`RESET_N`	I-D	FSM	Active-low reset signal for the CSI-2 TX block.
`ENABLE_TX`	I-D	FSM	Enables the CSI-2 transmitter operation.
`TX_MODE[1:0]`	I-D	FSM	Transmission mode selection: `00` = Standby, `01` = Low-Power (LP), `10` = High-Speed (HS).
`LANE_CFG[3:0]`	I-D	FSM	Configuration of active data lanes (e.g., 1 to 4 lanes).
`VIRTUAL_CH[1:0]`	I-D	FSM	Virtual channel identifier for multiplexing multiple data streams.
`DATA_TYPE[5:0]`	I-D	FSM	Specifies the data type (e.g., RAW10, RAW12, RGB) as per CSI-2 specification.
High-Speed Differential Outputs (to D-PHY)			

HS_CLK_P/N	O-D	PAD	Differential high-speed clock pair for D-PHY.
HS_DATA0_P/N	O-D	PAD	Differential high-speed data pair for lane 0.
HS_DATA1_P/N	O-D	PAD	Differential high-speed data pair for lane 1.
HS_DATA2_P/N	O-D	PAD	Differential high-speed data pair for lane 2.
HS_DATA3_P/N	O-D	PAD	Differential high-speed data pair for lane 3.
Low-Power Control Signals (per lane)			
LP_CTRL0	I-D	FSM	Low-power control signal for lane 0.
LP_CTRL1	I-D	FSM	Low-power control signal for lane 1.
LP_CTRL2	I-D	FSM	Low-power control signal for lane 2.
LP_CTRL3	I-D	FSM	Low-power control signal for lane 3.
Serializer Enable Signals (per lane)			
SER_EN0	I-D	FSM	Serializer enable for lane 0.
SER_EN1	I-D	FSM	Serializer enable for lane 1.
SER_EN2	I-D	FSM	Serializer enable for lane 2.
SER_EN3	I-D	FSM	Serializer enable for lane 3.
Test and Debug			
TEST_MODE	I-D	FSM	Activates test mode for internal diagnostics.
TEST_PATTERN[7:0]	I-D	FSM	Test pattern data input for generating known patterns during testing.
HS_MONITOR[3:0]	O-D	FSM	Monitors the status of high-speed data lanes for debugging purposes.
FSM_STATE[3:0]	O-D	FSM	Current state of the internal finite state machine (FSM) for status monitoring.
DEBUG_CLK_OUT	O-D	FSM	Provides internal debug clock output for test equipment synchronization.

Power and Ground			
VDD	P-D	PAD	Core digital power supply (e.g., 1.8 V).
GND	P-D	PAD	Digital ground reference.
AVDD_PHY	P-A	PAD	Analog power supply for the PHY and output drivers (e.g., 3.3 V).
AGND_PHY	P-A	PAD	Analog ground reference for the PHY.
BYP_BIAS	I-A	INT.	Bias voltage input for analog circuits (optional, depending on design).

8.1. CSI-2 Controller

The CSI-2 controller is a fully digital block responsible for receiving serialized pixel data—transferred one pixel at a time—from an upstream serializer. The controller formats these pixels into MIPI CSI-2 compliant packets by adding headers, payload length, virtual channel identifiers, and error-checking codes (ECC and CRC). Once a packet is complete, it is distributed across the active high-speed data lanes and prepared for transmission through the D-PHY physical interface.

The controller operates independently of the analog PHY. Therefore, signals related to differential transmission (HS_DATAx_P/N), low-power modes (LP_CTRLx), and physical enable controls (SER_ENx) are not part of the CSI-2 controller but are managed by the external PHY block.

The controller interfaces with the rest of the system through three main ports:

- **MIPI input interface**: handles ADC output data and line/frame synchronization flags.
- **Control and Configuration**: handles transmission modes, virtual channel selection, and pixel format.
- **PHY Output**: delivers serialized packet data and clock signals to the PHY block for high-speed transmission.

This modular split allows for a clean separation between data formatting (CSI-2 controller) and physical layer signaling (PHY), facilitating easier integration and testing.

CSI-2 Controller (Digital logic):

- Packetizes pixel data into CSI-2 protocol packets (frame, line, payload)
- Adds headers, CRC, and packet delimiters
- Manages lane distribution and timing
- Operates fully in digital logic (synthesizable in VHDL/Verilog)
- Interfaces with internal image sensor logic (pixel bus)

- **Tab. 8.2 MIPI CSI-2 Controller's external signals**

Signal Name	Direction	Description
Column-ADC Interface		
Pix_Clk	I-D	Pixel clock used to latch Pix_Data.
Pix_Data[0:15]	I-D	Parallel pixel data bus.
Pix_Valid	I-D	Indicates valid pixel data on Pix_Data.
Line_Valid	I-D	Indicates active video line
Frame_Valid	I-D	Indicates active video frame
Control and Configuration		
CLK_SYS	I-D	System clock for internal logic and configuration registers.
RESET_N	I-D	Active-low reset signal for the CSI-2 TX block.
ENABLE_TX	I-D	Enables the CSI-2 transmitter operation.
TX_MODE[1:0]	I-D	Transmission mode selection: 00 = Standby, 01 = Low-Power (LP), 10 = High-Speed (HS).
LANE_CFG[3:0]	I-D	Configuration of active data lanes (e.g., 1 to 4 lanes).
VIRTUAL_CH[1:0]	I-D	Virtual channel identifier for multiplexing multiple data streams (0-3).
DATA_TYPE[5:0]	I-D	Specifies the data type (e.g., RAW10, RAW12, RGB) as per CSI-2 specification.
Interface to D-PHY		
TX_CLK	D-O	Clock for PHY (used in HS mode)
TX_DATA_LANE0	D-O	Serialized CSI-2 packet data for lane 0
TX_DATA_LANE1	D-O	Serialized CSI-2 packet data for lane 1 (if enabled via LANE_CFG)
TX_DATA_LANE2	D-O	Serialized CSI-2 packet data for lane 2 (if enabled via LANE_CFG)

TX_DATA_LANE3	D-O	Serialized CSI-2 packet data for lane 3 (if enabled via LANE_CFG)
TX_SYNC_CTRL	D-O	Handshake/control signals for PHY sync (e.g. Start-of-Transmission)
Test and Debug		
TEST_MODE	I-D	Activates test mode for internal diagnostics.
TEST_PATTERN[7:0]	I-D	Test pattern data input for generating known patterns during testing.
HS_MONITOR[3:0]	O-D	Monitors the status of high-speed data lanes for debugging purposes.
FSM_STATE[3:0]	O-D	Current state of the internal finite state machine (FSM) for status monitoring.
DEBUG_CLK_OUT	O-D	Provides internal debug clock output for test equipment synchronization.
Power and Ground		
VDD	P-D	Core digital power supply (e.g., 1.8 V).
GND	P-D	Digital ground reference.

8.2. D-PHY TX

D-PHY TX (Mixed-Signal / Analog Front-End)

- Serializes the parallel data stream into 1–4 high-speed differential lanes
- Transmits High-Speed (HS) signals using analog current-mode drivers (LVDS-like)
- Handles Low-Power (LP) states for idle, escape, and control sequences
- Includes biasing circuits, slew rate control, and common-mode voltage regulation
- Usually implemented as a hard IP block or analog custom block

The analog part has the blocks summarized in Tab. 7.1.

Tab. 8.4 Output analog physical layer parts

Parameter	MIPI D-PHY (TX HS mode)	Notes
Differential swing	140 mV – 270 mV typ. (±70–135 mV)	MIPI uses a lower swing to reduce power
Common-mode voltage	0.8 V – 1.3 V (typ. 1.2 V)	

Voltage on HS+ or HS−	200 mV – 400 mV above GND	Depends on swing and termination
Termination impedance	100 Ω differential (at RX)	External or internal in receiver
LP (Low-Power) voltage	CMOS levels (0 V / 1.2–1.8 V)	Used during idle, reset, and transitions
TX current drive	~3–4 mA per lane (for 270 mV)	Controlled current-mode source

Tab. 8.5 Output analog physical layer parts

Signal Name	Direction	Description
High-Speed Differential Outputs (to D-PHY)		
HS_CLK_P/N	O-D	Differential high-speed clock pair for D-PHY.
HS_DATA0_P/N	O-D	Differential high-speed data pair for lane 0.
HS_DATA1_P/N	O-D	Differential high-speed data pair for lane 1.
HS_DATA2_P/N	O-D	Differential high-speed data pair for lane 2.
HS_DATA3_P/N	O-D	Differential high-speed data pair for lane 3.
Low-Power Control Signals (per lane)		
LP_CTRL0	O-D	Low-power control signal for lane 0.
LP_CTRL1	O-D	Low-power control signal for lane 1.
LP_CTRL2	O-D	Low-power control signal for lane 2.
LP_CTRL3	O-D	Low-power control signal for lane 3.
Serializer Enable Signals (per lane)		
SER_EN0	O-D	Data enable for lane 0.
SER_EN1	O-D	Data enable for lane 1.
SER_EN2	O-D	Data enable for lane 2.
SER_EN3	O-D	Data enable for lane 3.
Power and Ground		
VDD	P-D	Core digital power supply (e.g., 1.8 V).

GND	P-D	Digital ground reference.
AVDD_PHY	P-A	Analog power supply for the PHY and output drivers (e.g., 3.3 V).
AGND_PHY	P-A	Analog ground reference for the PHY.
BYP_BIAS	I-A	Bias voltage input for analog circuits (optional, depending on design).

8.2.1. Output lines operation

In a MIPI D-PHY interface, image and video data are always transmitted using the High-Speed (HS) mode, which employs differential signaling on each data and clock lane. The HS mode enables high-throughput, low-voltage differential transmission, suitable for continuous pixel streaming. In contrast, the Low-Power (LP) mode uses single-ended CMOS signaling on the same physical lines, but it is not used for image transfer. Instead, LP mode is reserved for link control, initialization sequences, and escape commands, such as entering standby, issuing soft resets, or performing low-speed communication. The PHY dynamically switches between HS and LP modes depending on the transmission state, but image data is exclusively handled in HS mode.

9. Integration reference

Regarding the chip's external connections, several passive components (both resistive and capacitive) are mandatory for its operation:

Specifically, it is necessary to connect:

1. A 120 Kohm resistor between the *Iref_RESOUT* and *GND* pins.
2. Two 47nF capacitors between the following pins:
   - *Cfly_CPpos_N* and *Cfly_CPpos_P*
   - *Cfly_CPneg_N* and *Cfly_CPneg_P*
3. Two 22uF capacitors connected to the *VDDH* and *VSSH* pins.

Fig. 9.1 ET2-GS-FC externa connections

10. Appendix A – data timing

In this section timing diagram to control ET2-GS-FC are included.

10.1. Integration/Readout Phase of the ET2-GS-FC Sensor

The first stage of the **ET2-GS-FC** sensor control concerns light signal acquisition. Like all integration-based sensors, the signal is captured during a defined integration period, during which the photodiode is left free to react to the incident light.

In the specific case of the pixel integrated into the **ET2-GS-FC** chip, the pixel consists of a pinned photodiode with global shutter. The key signals involved in the acquisition process, and synchronous to the master clock of the system (MCLK), are:

- GRST (Global Reset): This signal is typically used for antiblooming and for resetting the photosensitive area before starting the signal acquisition. In the case of the ET2-GS-FC sensor, GRST is dynamically controlled, as detailed in Section 5.1.1.1.
- Tx1 (Transfer Gate): This signal is responsible for transferring the charge from the photodiode to the floating diffusion (or memory node, see Section 5.1.1 related to the pixel structure). The charge transfer occurs simultaneously for all pixels, moving the stored signal from the photodiode to the floating diffusion, where it is held until readout.

Once the charge transfer is completed, the frame readout can begin. At the same time, a new integration cycle can start in parallel.

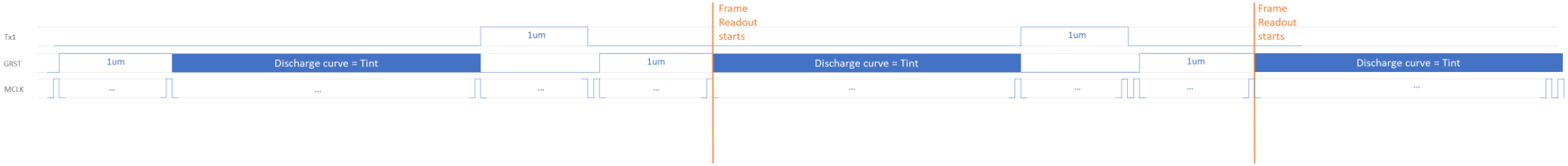


Fig. 10.1Integration and readout phases

It is crucial to ensure proper synchronization between the integration and readout phases, even when their durations do not match. Specifically, two scenarios must be considered:

- When the integration time is shorter than the readout time.
- When the integration time exceeds the readout time.

These cases require careful management to avoid artifacts and ensure optimal signal acquisition and processing.

Descrizione della temporizzazione

Fig. 10.2 Frame readout – row selection

Fig. 10.3 Column ADCs' readout

11. Appendix B – design rules notes

11.1. Notes for IO library:

Using a **TBE back-end**, pick the stack pad type and not the pol (pad over logic) type.

See location within our IO libraries: lib/dfii/oa/stack/* and /lib/gds/stack/*

Schematic views are not available, only layout and symbol views.

For LVS please include the cdl netlist (/lib/cdl/*.cdl) in the LVS run (see instructions in our following "Analog Physical Verification Usage Guide" on page 299.

As for Simulation, include the spice netlist in your simulator, see instructions on page 24 in "Analog Physical Verification Usage Guide" for ADE tool, see location of spice netlist within our IO libraries (/lib/spc_simple/[library name].spc (for pre-layout simulation), lib/spc/[library name].spc (for post-layout simulation, has parasitics)).

See explanation about std-cell and IO libraries options (section #6 from page 12) in the guide (TS18IS Design Kit Guide).

11.2. Supply strategy

Description	3.3V-Only	3.3V/1.8V	1.8V-Only
Single Power, Single Ground			
Power Supply for IO Side, Core Side and IC	PVDF_33_33	-	PVDF_18_18
Ground Supply for IO Side, Core Side and IC	PV0F_33_33	-	PV0F_18_18
Dual Power, Single Ground			
Power Supply for IO Side and IC circuits	PVDA_33_NA	PVDA_33_NA	PVDA_18_NA
Power Supply for Core Side and IC circuits	PVDI_NA_33	PVDI_NA_18	PVDI_NA_18
Ground Supply for IO Side, Core Side and IC	PV0F_33_33	PV0F_33_18	PV0F_18_18
Single Power, Dual Ground			
Power Supply for IO Side, Core Side and IC	PVDF_33_33	-	PVDF_18_18
Ground Supply for IO Side and IC circuits	PV0A_33_33	-	PV0A_18_18
Ground Supply for Core Side and IC circuits	PV0I_33_33	-	PV0I_18_18
Dual Power, Dual Ground			
Power Supply for IO Side and IC circuits	PVDA_33_NA	PVDA_33_NA	PVDA_18_NA
Power Supply for Core Side and IC circuits	PVDI_NA_33	PVDI_NA_18	PVDI_NA_18
Ground Supply for IO Side and IC circuits	PV0A_33_33	PV0A_33_18	PV0A_18_18
Ground Supply for Core Side and IC circuits	PV0I_33_33	PV0I_33_18	PV0I_18_18

Fig. 11.1 Available supply PADs

In this design, the pad ring is structured to ensure proper separation between analog and digital power domains while maintaining ESD protection and power continuity. The analog core is powered through dedicated analog power pads, specifically: PVDI_NA_18 for VDDA and PV0I_33_18.

To separate analog and digital power domains along the pad ring, cut cells are inserted for power supply (VDD/VDDA) with cell PCDI_NA_18. GND and GNDA will remain connected but back-to-back path is included to clean the analog ground (GNDA).

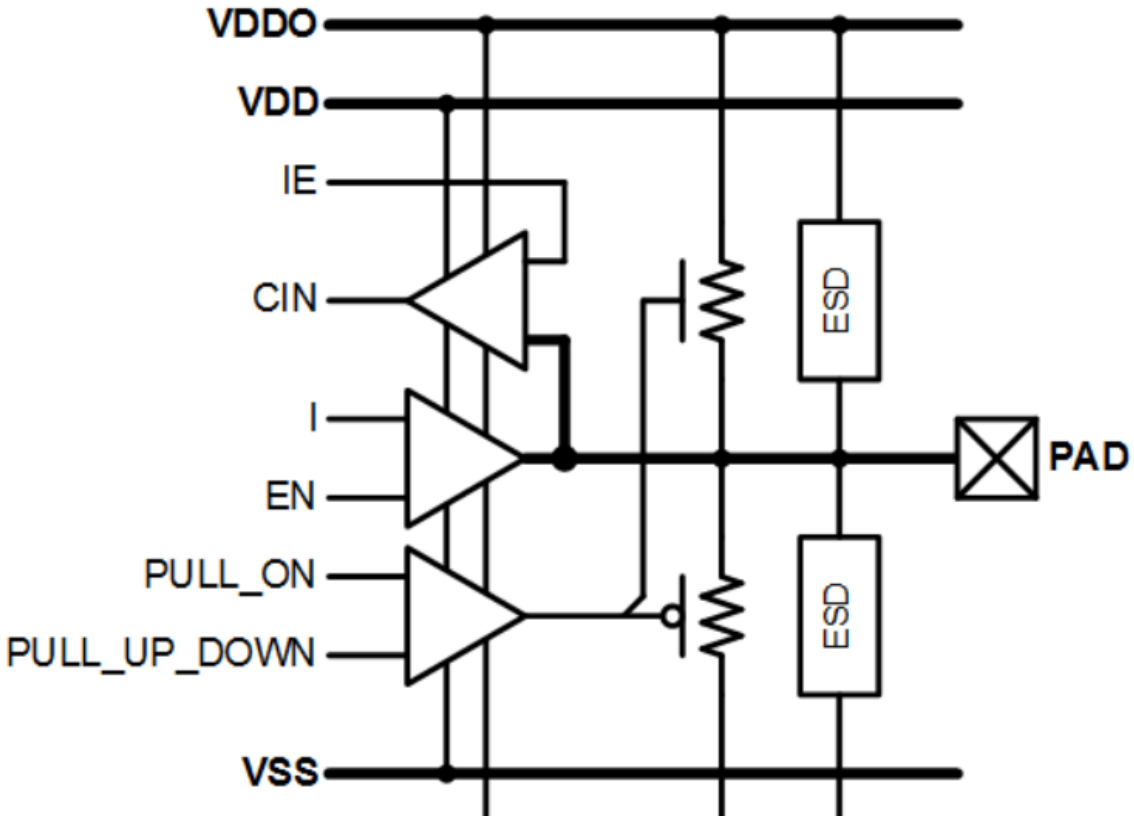
The I/O supply ring (VDDO/VSSO) is kept continuous across the full perimeter of the chip to ensure proper operation of all pads. Though analog pads also include VDDO/VSSO terminals, this is accepted practice, and these pads are allowed to share the I/O supply ring. In the analog region, care is taken in the layout to minimize current injection from digital I/O activity.

However, filtering and proper decoupling capacitors are implemented externally on the PCB to isolate sensitive analog blocks.

11.3. Digital PADs

PIOH is input/output data cell, with pull-up/down resistor, drive strength selector, Schmidt-trigger option, TTL/CMOS input levels and built-in ESD protection. The IO cell supports over-voltage tolerance up to VDDO level.

FUNCTIONAL SCHEME



PIN DESCRIPTION

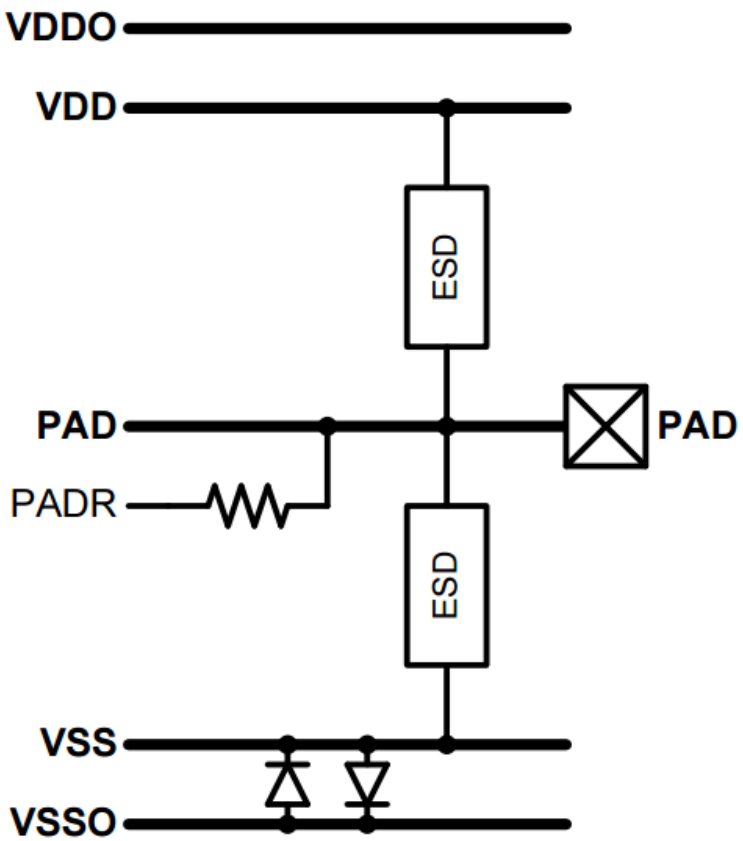
Pin	Type	'0'	'1'	Description
PAD	Digital Input	0V[a]	VDDO[a,b]	Bond pad
	Digital Output	0V[a]	VDDO[a]	
CIN	Digital Output	0V	VDD	Data in from PAD
I	Digital Input	0V	VDD	Data out to PAD
EN	Digital Control	0V	VDD	Output enable
IE	Digital Control	0V	VDD	Input enable
PULL_UP_DOWN	Digital Input	0V	VDD	Pull-up/pull-down resistor data
PULL_ON	Digital Control	0V	VDD	Pull-up/pull-down resistor enable
SCHM_ON	Configuration	0V	VDD	Schmidt-trigger: '1' – Enable / '0' – Disable
DRIVE_STR	Configuration	0V	VDD	Drive strength: '1' – 16mA / '0' – 8mA
TTL_ON	Configuration	0V	VDDO	Input levels: '1' – TTL / '0' – CMOS
VDD_OK	Supervision[c]	0V	VDD	'1' – Enable / '0' – Disable
VDDO_OK	Supervision[c]	0V	VDDO	'1' – Enable / '0' – Disable

FUNCTIONAL DESCRIPTION

Output Drive				
Inputs				Output
EN	**I**	**PULL_ON**	**PULL_UP_DOWN**	**PAD**
0	X	0	X	High-Z
0	X	1	0	Weak-0
0	X	1	1	Weak-1
1	0	0	X	0
1	1	0	X	1

Input Drive			
Inputs			Output
EN	**IE**	**PAD**	**CIN**
0	1	0	0
0	1	1	1
X	0	X	0

FUNCTIONAL SCHEME



PIN DESCRIPTION

Pin	Type	Voltage	Description
PAD	Analog Input	0V–VDD	Bond pad, suitable for transistors drains[a]
	Analog Output	0V–VDD	
PADR[b]	Analog Output	0V–VDD	Suitable for transistors gates

11.4. Analog PADs

For analog signals are available two possible PADs: PAL and PAH, with built-in ESD protection. The cells respectively support over-voltage tolerance up to VDD and VDDO levels. In the analog part of the ring, score and PAD supplies will be at the same level (3.3V).

11.5. CUT cells

Description	3.3V-Only	3.3V/1.8V	1.8V-Only
Cut Cells			
Cut of IO Side Power Supply	PCDA_33_NA	PCDA_33_NA	PCDA_18_NA
Cut of Core Side Power Supply	PCDI_NA_33	PCDI_NA_18	PCDI_NA_18
Cut of IO Side Ground Supply	PC0A_33_NA	PC0A_33_NA	PC0A_18_NA
Cut of Core Side Ground Supply	PC0I_NA_33	PC0I_NA_18	PC0I_NA_18
Coupling (Diode) Cells			
Coupling of IO Side Power Supply	PDDA_33_NA	PDDA_33_NA	PDDA_18_NA
Coupling of Core Side Power Supply	PDDI_NA_33	PDDI_NA_18	PDDI_NA_18
Coupling of IO Side Ground Supply	PD0A_33_NA	PD0A_33_NA	PD0A_18_NA
Coupling of Core Side Ground Supply	PD0I_NA_33	PD0I_NA_18	PD0I_NA_18

Fig. 11.2 Available CUT and COUPLING Cells

While PCDI_NA_18 will be used to separate VDD and VDDA, PD0I_NA_18 will be used for cleaning the GNDA supply.